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# CHAPTER 1

# INTRODUCTION

Plastic pollution is one of the most pressing environmental challenges of our time. Mainly derived from finite petrochemical resources, globally, 400 million tons of plastic are produced annually, of which only a small fraction of 14% is recycled effectively. This results in massive waste accumulation, with million tons of plastic entering natural ecosystems (1). Addressing these challenges, biodegradable polymers from renewable resources have emerged as promising alternatives. Biopolymers are polymers produced by natural sources either artificially integrated from a natural material or completely biosynthesized by living organisms. There are various biopolymers available, but focus is laid only on four basic biopolymers based on comparatively better properties and which are most used as matrix materials. The following are the polymers with abbreviations.

- Poly lactic acid or poly lactide - PLA
- Poly glycolic acid or poly glycolide - PGA
- Poly- ε-caprolactone or poly caprolactone - PCL
- Poly hydroxyl alkanoates - PHA

A lot of these polymers have good film characteristics that make them viable for applications in high performance applications and also for traditional commodity uses. Other applications of these polymers can be in food containers, soil retention sheeting, agriculture firm, waste bags and as packaging material. There are huge applications in the medical field (2).

Composite materials are multiphase systems obtained by the artificial combination of at least two chemically distinct materials, separated by a well-defined interface, in order to achieve superior performance compared to the

individual component materials. Alternatively, a composite material can be defined as a material composed of multiple components, whose organization and formulation allow for the exploitation of the best characteristics of each component, creating a material with final properties that exceed those of its base components. Bio composites, a subcategory of composite materials, represent a modern and sustainable solution. They utilize natural fibres and biodegradable matrices to produce durable materials that address contemporary challenges related to sustainability and reducing carbon footprints. Natural fibres, such as flax, hemp, or bamboo, provide bio composites with mechanical strength and rigidity, while biodegradable matrices (natural or synthetic polymers) ensure the cohesion of the material and contribute to its ecological properties. Unlike traditional composites made from synthetic fibres and petroleum-based matrices, bio composites can be integrated into a circular life cycle, being recyclable or compostable. In recent years, bio composites have found applications in various industries, including construction, automotive, medical equipment and consumer products. These materials are valued for their unique combination of lightness, durability and sustainability attracting the interest of researchers and developers aiming to expand their use and technological innovation in production.

*Chromolaena odorata* is one of the herbs that belongs to *Asteraceae*, the sunflower family, and is also known as *Eupatorium odoratum*. It acts as a traditional medicinal plant. Several parts of this plant are widely used to treat wounds, burns, skin infections as well as to possess anticancer, antidiabetic, anti-hepatotoxic, anti-inflammatory, antimicrobial, and antioxidant properties. *Chromolaena odorata* is also considered a serious weed in plantation crops in the world, such as palm oil plant, coconut, rubber and citrus. It consists of high allelopathic compounds and suppresses neighbouring vegetation. This weed affects

particularly small farms, production in agricultural sectors and as well as natural ecosystems. The proximate composition of *Chromolaena odorata* revealed that the leaves contain protein, fibre, carbohydrate and moisture. The phytochemical analysis of *Chromolaena odorata* leaves showed that it includes steroids, saponins, alkaloids, flavonoids, phytates, cyanogenic glycosides, tannins, oxalates, anthocyanins, and phenols (3)(4).

Poly Lactic Acid is a biodegradable polymer economically produced from renewable resources and can be used in compostable products. PLA is not as important as traditional commodity polymers like PET or PVC. Its widespread application has been hindered by numerous physical and processing short comings. PLA is the most widely used plastic filament material in FDM 3D printing, due to its low melting point, high strength, low thermal expansion and good layer adhesion, although it possesses poor heat resistance unless annealed. Poly lactic acid (PLA)-based materials are among the most promising bio plastics for food packaging because they are biodegradable, derived from renewable resources and suitable for both flexible and rigid applications(5).

The combination of *Chormoleana oderata* leaves extract and PLA films propose a versatile platform for developing sustainable and multifunctional materials with multiple applications and alternative solutions to conventional petrochemical plastics. Therefore, this study focuses on PLA- *Chromolaena odorata* leaves extract biofilms. *Chromolaena odorata* leaves extract incorporated PLA films were prepared at room temperature by a simple film casting technique. Further, the IR spectrum and colour adsoption properties of films were studied.

# CHAPTER 2

# REVIEW OF LITERATURE

A polymer is a material made up of extremely large molecules called macromolecules, which are formed by linking many repeating units known as monomers. Because of their wide range of characteristics, both natural and synthetic polymers are widely used and play an important role in daily life. Their large molecular size, compared with small molecules gives them distinctive physical properties such as strength, elasticity, viscoelastic behavior and the ability to form amorphous or semi crystalline structures instead of well-defined crystals**.** The study of polymers is an important part of polymer science, which includes polymer chemistry and polymer physics, as well as materials science and engineering (6). The properties of polymers are determined by their structure and they are categorized into different classes based on their physical characteristics. Various physical and chemical properties explain how a polymer behaves as a continuous macroscopic material. According to thermodynamic principles, these properties are grouped into bulk properties and intensive properties. In modern times, polymers have become extremely important and are used in almost every field of daily life**.** But synthetic polymers, especially plastics, pose serious environmental problems because they are non-biodegradable and can remain in nature for hundreds of years. They accumulate in soil and water bodies, leading to pollution and harming ecosystems. In oceans and rivers, plastic waste can injure or kill marine organisms through ingestion or entanglement. In addition, many plastics contain toxic chemicals such as BPA and phthalates that may leach into the environment. When plastics are burned, they release greenhouse gases and hazardous fumes, which contribute to air pollution and climate change. We are living in a time when plastic pollution is increasing at an alarming rate. Plastics are

widely dispersed throughout the environment, posing a serious threat to ecosystem balance and human health. Apart from their physical persistence, plastics can also act as carriers of harmful substances such as pesticides, polyaromatic hydrocarbons, diphenyl compounds and pharmaceutical residues. Their extensive use in everyday items, including packaging materials and water bottles has significantly increased plastic waste. Therefore, there is an urgent need to develop biodegradable or recyclable plastics, which can help reduce dependence on petroleum resources, lower $CO_2$ emissions and decrease the amount of waste that must be disposed of.

Recent progress in biomaterials**,** especially in the development of biodegradable polymers, along with advanced bio-fabrication technologies, has significantly increased interest in biopolymers among consumers and industrial sectors. These materials are now being used in a wide range of applications, from plastic packaging products to medical devices. Consequently, industries are actively searching for sustainable production methods to satisfy the rising demand for environmentally friendly polymers. In this context, poly lactic acid (PLA) has emerged as a promising and sustainable alternative (7). PLA is aliphatic polyester composed of blocks of lactic acid or 2- hydroxyl propanoic acid. The building block for the synthetic PLA polymer is lactic acid, which was isolated the first time by the Swiss chemist Scheele in the 1700s and its first biomedical application was in the repair of mandibular fractures in dogs (8). However, because of its high cost and low availability, the use of PLA was firstly limited to medical applications, but currently high molecular weight PLA can be processed by extrusion moulding, injection moulding, blow moulding, electrospinning and thermoforming. PLA also offers similar thermal, mechanical, optical and barrier properties compared to commercially available polymers, such as polystyrene (PS), polypropylene (PP), and polyethylene terephthalate (PET), thus widening its use for a range of other

applications (8). Even though PLA exhibits several useful properties, its brittleness, susceptibility to hydrolysis, low impact resistance and low elongation at break are some of the limiting factors for widespread commercial application of this material. In order to overcome these deficiencies, several methods were developed by various researchers in this field and some of them are: i) blending with plasticizers, ii) copolymerization, iii) blending with other polymers, iv) preparing nano composites and v) surface modifications (10). Changing into polymer blends and composites can be used as a performance improvement technique for PLA. The main attraction for the development of blends of PLA with other biodegradable materials is the improvement of properties of PLA without compromising the biodegradability of the resulting blend system. PLA Composites can be prepared by PLA with starch, PLA and natural fibres, PLA with Poly (ε-caprolactone) (PCL), PLA with Poly hydroxybutyrate (PHB), PLA with Polyvinyl alcohol (PVOH) etc. Natural fibre reinforced composite (NFRC) was used in products, for instance, in the aerospace field, marine, transportation and defense. NFRC is unique because of its strength and stiffness properties. NFRC is a material that is composing fiber as a reinforcement that bound together by a matrix that has a significant influence in terms properties (11).

This study focus to develop and assess a PLA-based composite film infused with *Chromolaena odorata* leaves extract. The specific goals include synthesis and testing the same for dye adsorption and IR characterization.

# CHAPTER 3

# SCOPE OF THE RESEARCH PROBLEM

The interest in using natural fibres such as different plant fibres and wood fibres as reinforcement in polymers has increased dramatically during last few years. With regard to the environmental aspects it would be very interesting if natural fibres could be used instead of glass fibres as reinforcement in some structural applications. Natural fibres have many advantages compared to glass fibres, for example they have low density. Additionally they are renewable raw materials and have relatively high strength and stiffness. Their low-density values allow producing composites that combine good mechanical properties with a low specific mass. PLA composites are also suitable in clothing, disposable garments, upholstery, diapers and similar products. In these cases, PLA is processed into fibres or non-woven fabrics. For more than three decades, DuPont (E.I. du Pont de Nemours and Company, Wilmington, DE, USA has been developing materials and techniques based on two primary types of sheets: surface-bonded sheets, which are more rigid and resemble paper in texture and spot-bonded sheets, which are softer and more fabric-like. PLA composites have also found applications in everyday products like furniture, suitcases, grinding tools and automotive components, where enhancing mechanical and thermal performance is a key goal. In recent decades, bio-based polymer composite materials have been developed utilizing sustainable, eco based and green manufacturing concepts. The high reinforcing characteristics of cellulose, natural fibres, agricultural wastes, make it a valuable natural resource for incorporation as a raw material in polymer composites. *Chromolaena odorata* is primarily recognized for its medicinal, agricultural and environmental applications rather than as a traditional source of fibre. However it is an invasive seed with significant cellulosic contents and its stem and leaves have

been used as a natural filler or reinforcement in bio composites. Dye molecules are one of the most hazardous compounds for human and animal health, the excess intake of these materials can create toxic impacts. Several studies showed the practicality of the adsorption process for dye uptake from wastewaters. In recent years, various adsorbents were used to be efficient in this process. Among all, polymeric adsorbents demonstrate great applicability in different environmental conditions and attract many researchers to work on them. Hence synthesis of new composites and analysis of their dye adsorption capacity will be successful. In this work composites were fabricated using *Chromolaena odorata* leaves extract as filler and poly lactic acid as matrix and testing them for dye adsorption

# CHAPTER 4

# MATERIALS AND METHODS

## 1. *Chromolaena Odorata* leaves extract preparation

The collected *Chromolaena Odorata* leaves were washed several times with distilled water and then dried at room temperature. The dried leaves were grinded with an electric mixer grinder until it became fine powder. The fine powder obtained was preserved in bottles.

*Chromolaena Odorata* leaves powder weighing around 25 g was extracted in water for 8 hours using a 250 ml RB flask and leibig condenser. The leaves that dried, grinded and finely powdered is put inside the round bottom flask with added some pieces of porcelain tiles. Now the tube is closed with a piece of cotton above the condenser tube. The solvent water is added to the plant sample in the bottom flask which is then heated. Now the condenser evaporates and passes through the condenser where it condenses. Now the entire process is continuous until the extraction is completed. Extract were filtered through a Whatmann no- 40 filter paper.

## 2. Preparation of PLA films

Poly lactic acid (PLA), commercial-grade was purchased from Srushti Global Pvt. Ltd. India. 2 g of PLA granules were gradually added in 25 ml of chloroform with continuous stirring for 2 hours. The resulting mass was cast onto a watch glass and allowed to dry at room temperature for 24 hours. Once dried, the films were carefully peeled from the watch glass.

## 3. Preparation of PLA-COL films

PLA- *Chromolaena odorata* leaves (COL) extract composite films were prepared using solvent casting technique. Initially, 2 g of PLA granules were gradually added in 25 ml of chloroform and 1ml of extract was added and

continuous stirring for 2 hours. Three different films were prepared by changing the concentration of extract (3ml, 5ml). PLA and composite film was subjected to FTIR analysis.

## 4. Adsorption studies

One gram of Eriochrome black- T weighed and made up it into 100ml solution. Absorbance of the made up solution was measured. Using a pipette, 20 ml of Eriochrome Black T solution was transferred into different beakers. PLA and composites were added into different beakers. Shaken well and kept it for 24 hours. After 24 hours the solutions were filtered and measured the absorbance at 620 nm using colorimeter.

# CHAPTER 5

# RESULTS AND DISCUSSIONS

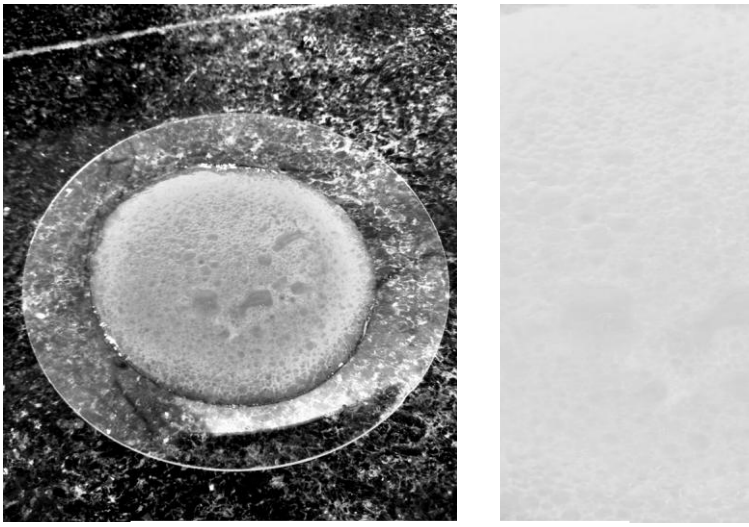

**Fig 1: PLA film**

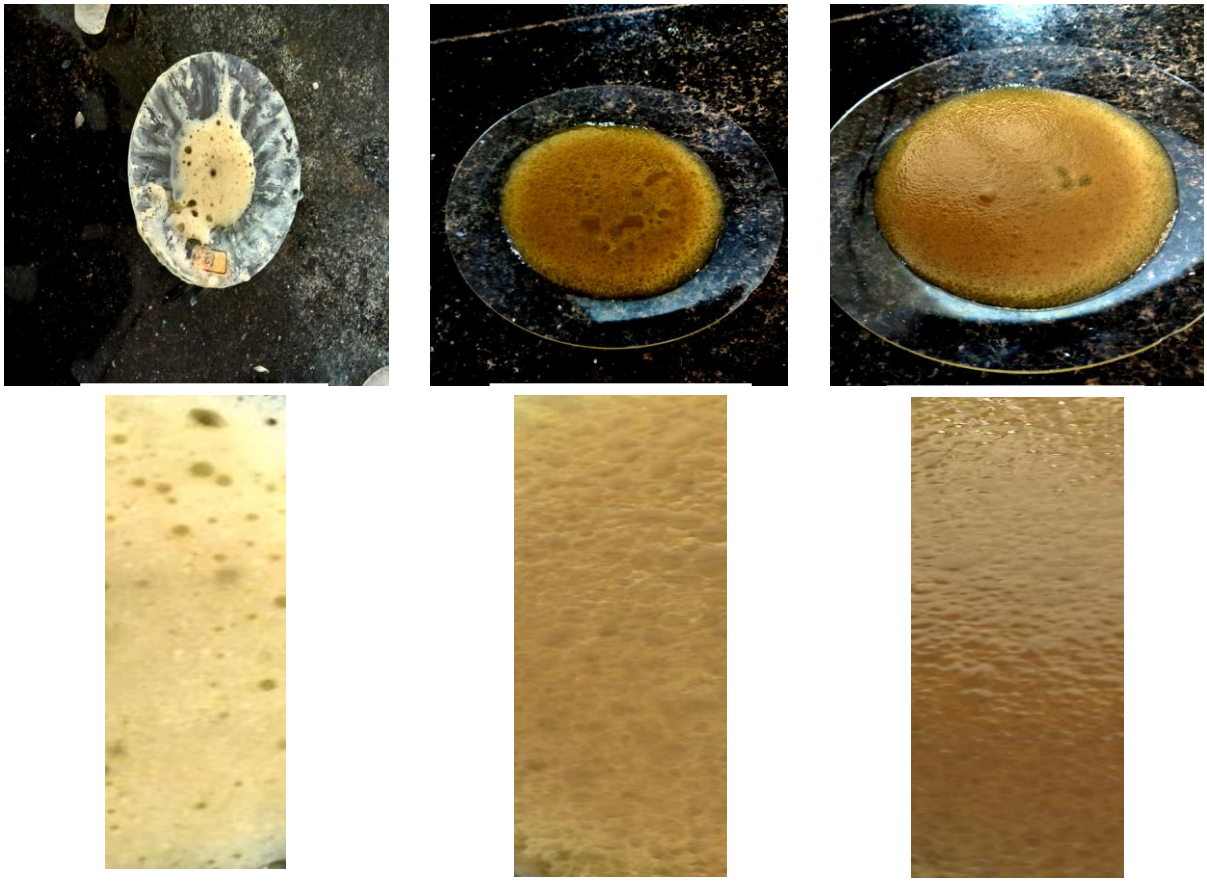

**Fig 2: PLA-COL FILMS**

**Preliminary Characteristics**

Colour: PLA Film- Off White

Composite – Pale Brown

Texture: Rough texture

Transparency: Opaque

Odour: No odour

**FTIR Analysis**

In this project work FTIR analysis was done from Central Instrumentation Facility, Pocker Sahib Memorial Orphanage College, Tirurangadi

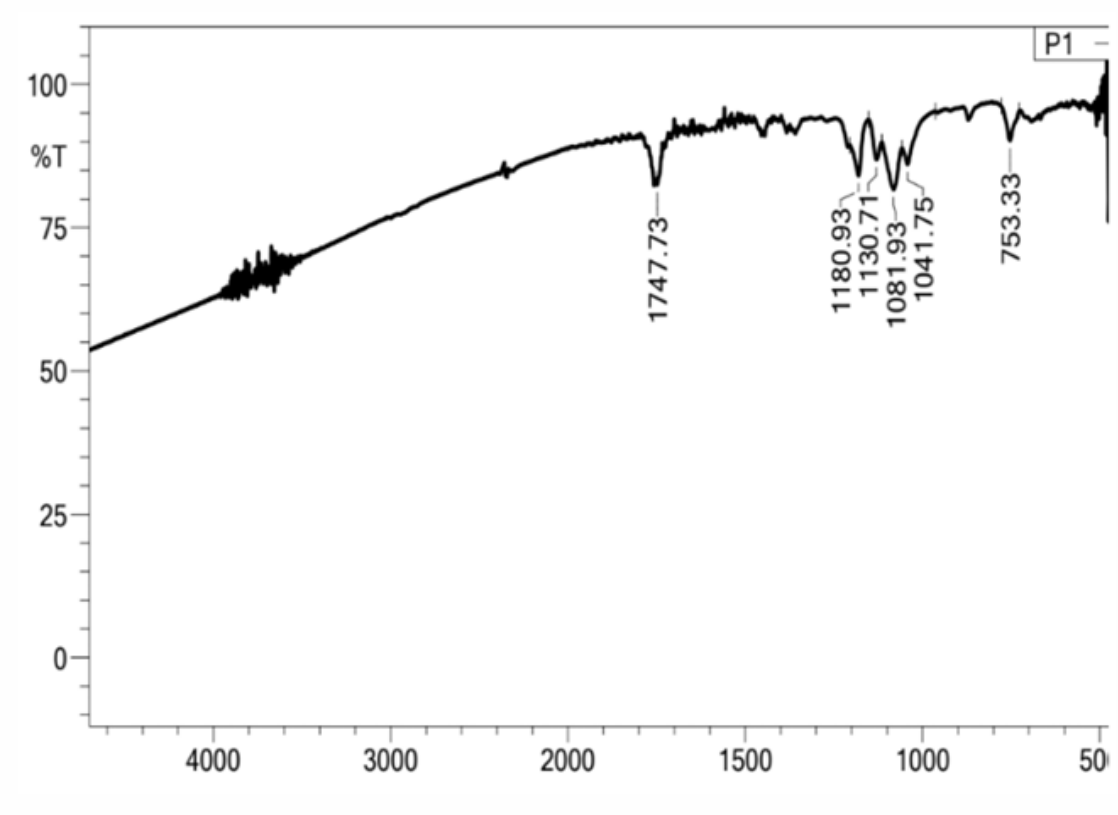


**Fig 3: FTIR Spectra of Poly Lactic Acid (PLA)**

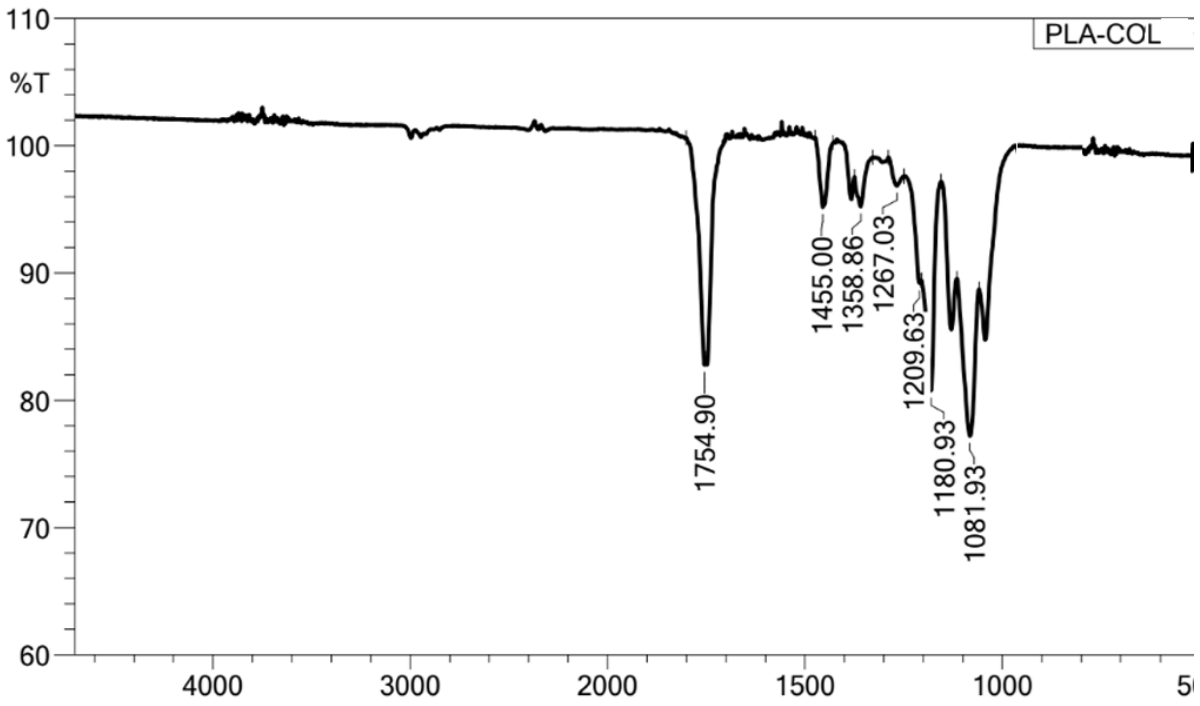


**Fig 4: FTIR Spectra of PLA-COL Composites**

Fig. 3 shows the FTIR spectra of poly lactic acid. The analysis findings established that C=O stretching band were responsible for the peak 1747.73 $cm^{-1}$. The absorption at 1180.93 $cm^{-1}$ is due to the ester carbonyl stretches, a characteristic of PLA's ester linkages. The multiple peaks at 1130.71 $cm^{-1}$ and 1081.93 $cm^{-1}$ indicate C–O–C stretching (ester bond), confirms the presence of ester linkage in PLA. The small peak at 1041.75 $cm^{-1}$ also corresponds to C–O stretching. This region is typically congested, with several overlapping signals frequently observed. The small to medium peaks observed in the 753.33 $cm^{-1}$ ranges are attributed to C–H bending or out-of-plane deformation. In the 3600 $cm^{-1}$, 3200 $cm^{-1}$ region, open O–H stretch is observed, suggesting the absence of free hydroxyl groups, moisture, or alcohols. FTIR spectra of PLA-COL composite film is shown in Fig.4. The intensity peaks at 1209.63 $cm^{-1}$, 1180.93 $cm^{-1}$, 1081.93 $cm^{-1}$ are corresponds to C=O stretching in ester group vibration. A moderately weak absorption bands at 1358.86 $cm^{-1}$ and 1455 $cm^{-1}$ are observed, corresponding to C–H bending in alkane group contributions. Furthermore, these peaks may also arise from aromatic amine compounds present in the *Chromolaena odorata* leaves extract, suggesting successful integration of plant-based materials into the PLA matrix. One unique

peak is seen in the pure PLA and the same peak moved in the composite films. The peak was seen at 1747.73 $cm^{-1}$ in the PLA spectra was moved to 1754.90 $cm^{-1}$ in the composite films.

**Colorimetric Analysis**

| SI No. | Samples | Absorbance |
|---|---|---|
| 1 | PLA | 1.31 |
| 2 | PLA-COL-1 | 0.85 |
| 3 | PLA-COL-2 | 0.62 |
| 4 | PLA-COL-3 | 0.22 |

# CHAPTER 6

# CONCLUSIONS

Natural and synthetic polymers showed significant adsorptive properties when combined with different adsorbent materials. Polymers in a composite structure showed good mechanical performance and additional binding sites for adsorption of foreign materials. In this study, the PLA-COL composites were prepared by the altering the concentration of *Chromolaena odorata* leaves extract. The resulting composites were characterized by FTIR. The adsorption of Eriochrome Black T dyes onto these composite were studied using colorimetry. The adsorption studies indicate that the adsorption increases as the amount of extract increases. The colorimetric method confirmed the colour adsorption on composites. Furthermore this composite is a low-cost material for the removal of Eriochrome black T from water, although advanced studies needed.

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